Kevin Ding

Vancouver, British Columbia | dingk3608@gmail.com | (587) 500-1960 | linkedin.com/in/dingk

Technical Skills

Electrical: THT/SMT Soldering, Multimeters, Oscilloscope, Circuit Analysis, PCB Design, Microcontrollers (Arduino, ESP32, STM32, EFM8)
Software: MATLAB, Altium Designer, Microsoft Office Suite, Git and GitHub, Quartus, ModelSim, LTspice
Languages: Assembly (8051, MIPS, ARM), SystemVerilog, C/C++, Python, HTTP

Education

University of British Columbia Vancouver, BC
Bachelor of Applied Science - Electrical Engineering (Year 3) *April 2027*
Relevant Courses: ELEC 202 - Circuit Analysis II | ELEC 291 - Electrical Design Lab I | ELEC 211 - Engineering Electromagnetics | CPEN 211 - Computing Systems I | CPSC 259 - Data Structures and Algorithms

Technical Work Experience

FUSE 33 Calgary, Alberta
Technician *May 2024 – August 2024*

- Prototyped an ESP 32-based embedded access control system for over 25 pieces of equipment communicating to a web server via HTTP.
- Designed an FSM-based access control algorithm in C++ using NFC technology, ensuring modularity for future updates.
- Collaborated with fellow students, utilizing Git for version control to maintain smooth project workflow.
- Taught classes on robotics and basic programming for the Arduino platform.

Engineering Student Teams

UBC Rocket University of British Columbia
Liquid Rocket Avionics *September 2022 – Present*

- Designed and soldered an SMT PCB with Altium to control parachute reefing cutters using an Arduino Nano MCU and barometric sensor.
- Assisted in building a parachute for supersonic deployment in high-altitude flight.
- Built a flight control PCB based on THT soldered parts to log flight data and control a Hydrolox engine

Technical Projects

Remote Controlled Car February 2024 – April 2024

- Collaborated on a remote-controlled car project with STM32, incorporating accurate steering and speed control.
- Utilized ARM, Python, and C/C++ to design a control system for a remote-controlled car.
- Used inherent electromagnetic behaviors of metals to create a metal detector circuit.
- Designed a motor control circuit and developed firmware, enabling precise remote operation.

Turing Complete Processor October 2023 – December 2023

- Designed an 8-bit Turing Complete processor with SystemVerilog and Quartus.
- Validated processor using test benches with ModelSim to verify functionality.
- Utilized a simplified ARM architecture to develop understanding of hardware and firmware design around RISC CPUs.

Robotic Arm May 2021 – Present

- Collaborated on a robotic hand using Arduino, integrating flex sensors and planning EMG system incorporation.
- Designed a radio operated remote control system capable of mimicking the motions of a human hand using flex sensors attached to a glove.

Other Work Experience

Kins Farm Market Coquitlam, British Columbia
Sales Associate *May 2023 – July 2023*

- Responsible for maintaining and improving customer experience, driving sales towards increasing profitability and fostering a positive/productive working environment.
- Worked collaboratively to continuously maintain and stock the store while managing a rotating inventory.
- Maintained a healthy and safe environment for customers and employees by adhering to proper food safety handling procedures.